Lesson 9:

Configuring Alarms



IGSS

Interactive Graphical
SCADA System

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Contents of this lesson



Topics:

- **Pre-defined alarms**
- **Creating alarms**
- **Alarm display on diagrams**
- **The Alarm list**
- **The Embedded Alarm list**

Predefined alarms

No.	Alarm text	Comment
1	Alarm in referenced diagram	Used for ”alarm see-thru” effect. A button / field referencing a diagram will blink if object on diagram enters alarm state.
10-11 20-21 30-51	General communication errors	Interface or PLC system errors. Apart from these, a set of error codes exist for each PLC driver (reported under the object ”Driver”)
90	Alarm generated by event	Appears if an event has been marked with the “Alarm on event” option.
91	Alarm Inhibit (All Alarms)	Appears if operator inhibits all alarms.
92	Alarm Inhibit (Specific Alarms)	Appears if operator inhibits a specific alarm on an object.
93	WinPager Inhibit	Appears if operator inhibits WinPager processing of an alarm

The pre-defined alarms are present in every IGSS project and must ***never*** be used for ordinary alarm purposes – they are reserved for system and communication errors.

Predefined alarms (cont.)

No.	Alarm text	Comment
94	Disk usage above first mark	The ”High Limit” defined for the ”System” object is exceeded. Default value is 80 % of harddisk used.
95	Disk usage above second mark	The ”High Alarm” defined for the ”System” object is exceeded. Default value is 90 % of harddisk used. Note: All writing to the harddisk is suspended when this alarm occurs, that is, no log files (*.log) or base class files (*.bcl files) are saved.
96	Object note attached	An operator has created an object note.
97	Winpager communication error	Warns if communication lost between Winpager and IGSS
98	Event	This alarm number is used for all events. When events are printed on the alarm printer, this alarm number occurs.
99	Undefined alarm received	
100	Last predefined alarm text	Alarms from number 101 may be used by the system designer. The limit is 10,000,000. The first 100 alarm numbers are reserved for system use.

Creating alarms



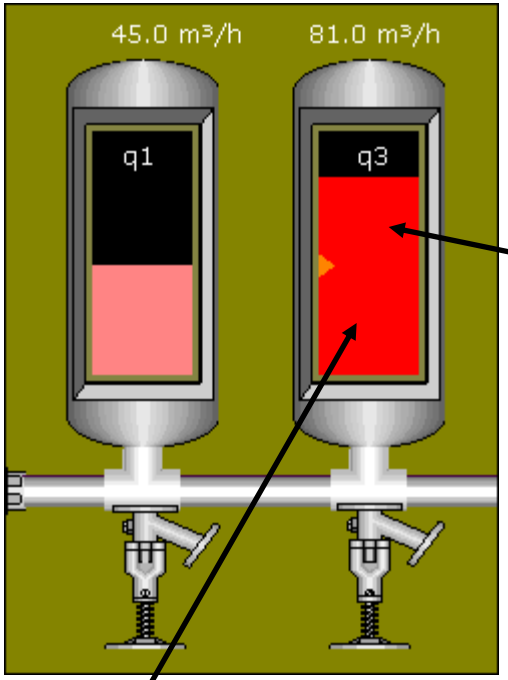
- Alarms can be created for analog, digital, counter and table objects
- The alarm text can be created in the properties dialog of the object itself or
- From the complete alarm text list (**Alarm Texts** in the **Edit** menu in **Definition**)
- The alarm atom must be selected, an alarm text must be created or selected and mapped to the alarm number
- Alarms with a higher priority number are considered more severe than ones with a lower priority number

Alarms and I/O modes

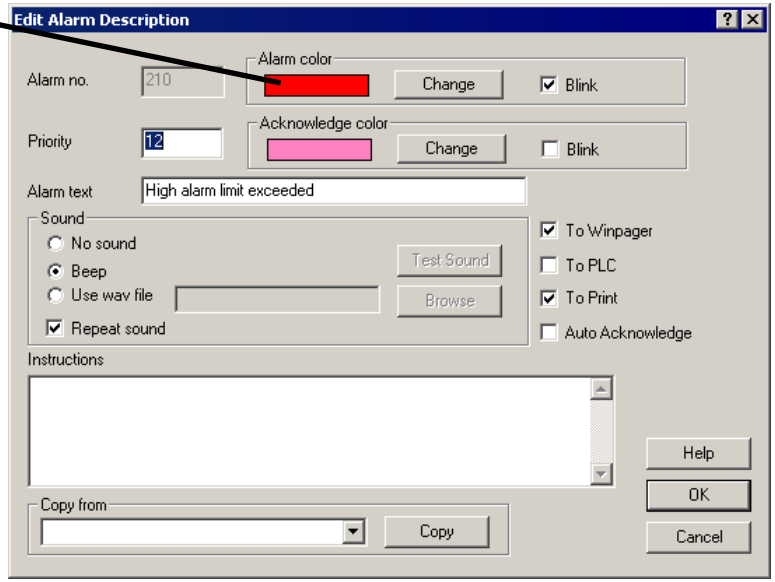


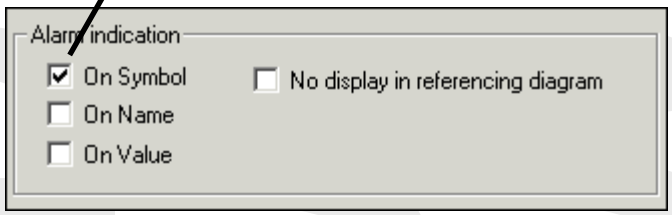
- The alarm atom can be set to **local** (which means the alarm is not issued from the PLC but is generated internally in IGSS)
- The alarm atom can be set to **in** (which means the alarm bit comes from the PLC)
- The alarm atom can be set to **out** (which means the alarm bit is sent back to the PLC as aknowledgement)

Alarm indication on diagram



Alarm text
Defines alarm and acknowledgment colors. Blink is typically enabled for alarm color but disabled for acknowledgment.



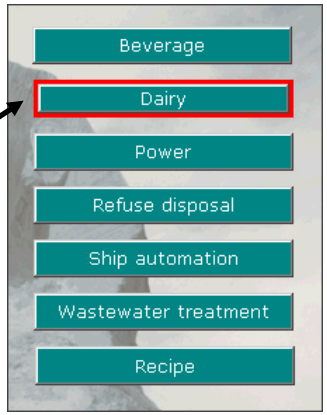


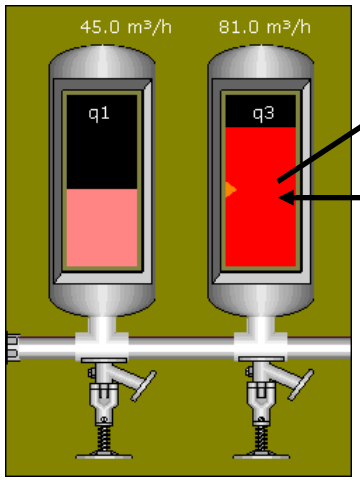
Display tab in object properties dialog box: settings for where to show the alarm on the object

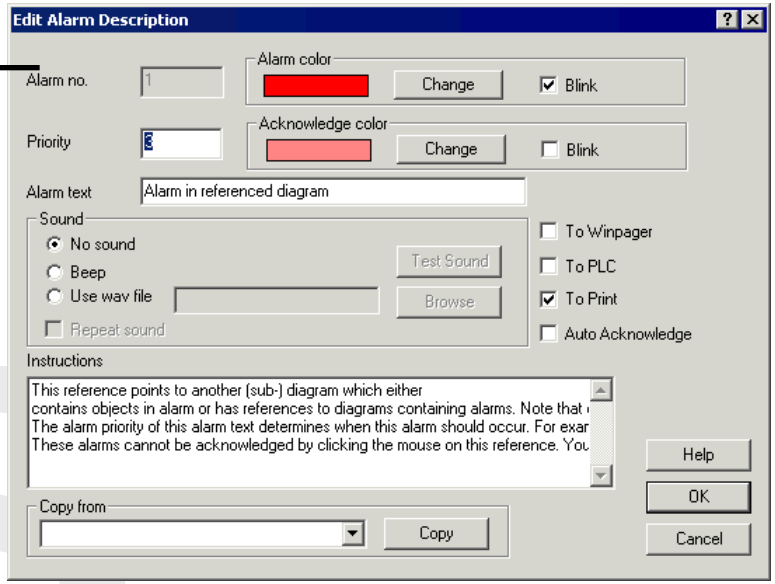
Alarm ”see-thru”

When an object on the **Dairy** diagram (where q3 is located) goes in alarm, the button on the overview diagram blinks according to the properties inherent in alarm text number 1.

NOTE The button must have at least a one pixel border around it for the alarm indication to be visible.





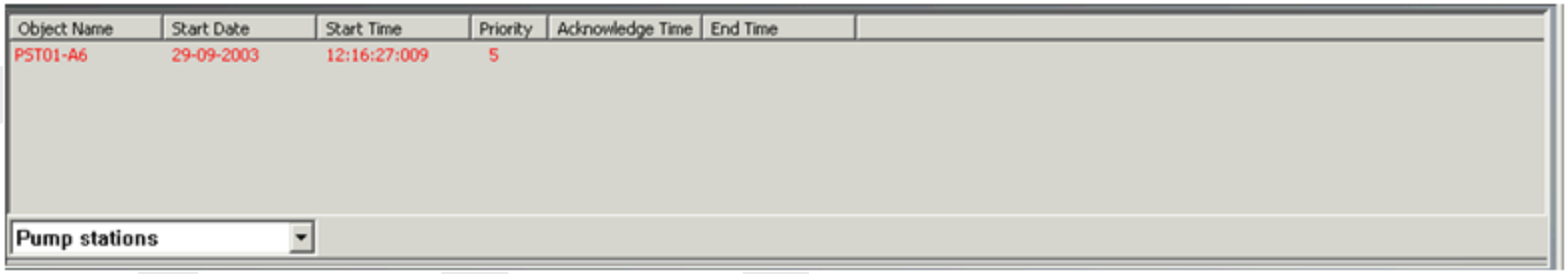


What is the Embedded Alarm List?

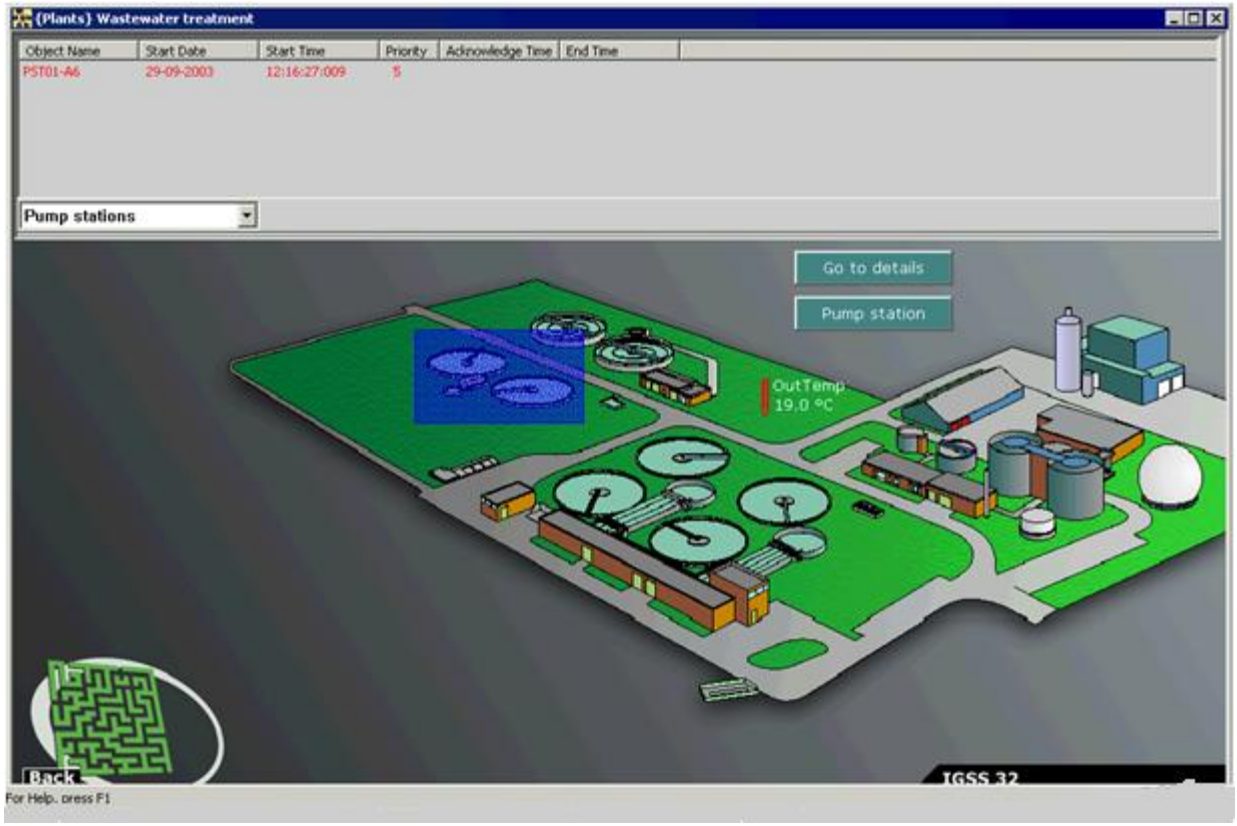
- a scaled down edition of the full alarm list
- gives the operator a quick overview of alarms from underlying diagrams
- usually anchored on an overview diagram – cannot be moved
- alarm filters are created for displaying specific alarms

Here's what it looks like

An embedded alarm list with a filter to only show alarms from **Pump stations**.



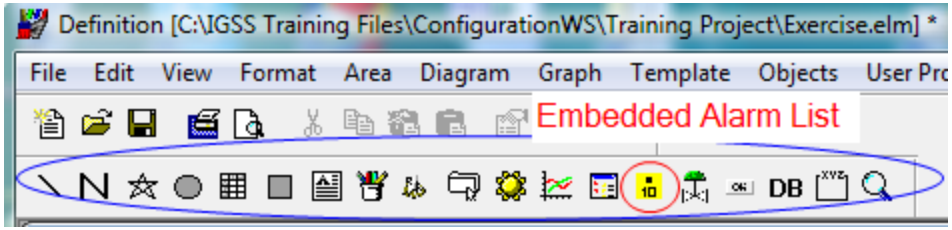
Anchored on overview diagram



Where do I find the Embedded Alarm List tool?

In the **Definition** module:

- Go to the menu and under **View** click on the **Drawing Tool Bar...**
- and its icons appear under the **Standard Toolbar**.
- Find the yellow alarm icon, click on it and drag it on to the diagram.



- Right click inside the resulting square and begin to select settings.

More information ...



Find presentations and exercises on www.7t.dk/igss